

Literature Evidence



Literature Basis

Ba-Y-Si-O family

Ba-Zn-Si-O structural neighbors

Y-Si-O process reference



Structural Hypothesis

Local composition grid

Zn-Y-O stoichiometric coupling

Competing phases, glass region



Screened Precursors

BaCO3 + Y2O3 + SiO2

ZnO / MgO / Co3O4

Model-ranked candidates

Ranking ≠ validation

Update composition and process

Experimental Items



Solid-State Route

Calcine



Regrind



Anneal

Bulk phase purity



Solution Growth

Flux



Dwell



Slow cool

Single-crystal structure

Result Feedback



Characterization

PXRD / Rietveld

SCXRD

EDS / EPMA / ICP

HT vs cooled phases

Open Questions

Does Zn enter and couple with oxygen stoichiometry?

Stable phase field, or liquid selectivity?

How do actual composition and structural signals correspond?